



Red Hat OpenShift AI Self-Managed 3.4

Working with AutoML

Use AutoML in Red Hat OpenShift AI Self-Managed

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Abstract

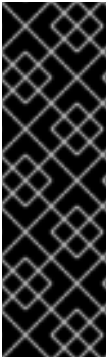
Use AutoML in Red Hat OpenShift AI Self-Managed to automatically train and compare machine learning models for your tabular data.

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PREFACE

You can use AutoML in OpenShift AI to train, evaluate, and compare machine learning models for your tabular data.



IMPORTANT

AutoML is a Technology Preview feature only. Technology Preview features are not supported with Red Hat production service level agreements (SLAs) and might not be functionally complete. Red Hat does not recommend using them in production. These features provide early access to upcoming product features, enabling customers to test functionality and provide feedback during the development process.

For more information about the support scope of Red Hat Technology Preview features, see [Technology Preview Features Support Scope](#).

CHAPTER 1. AUTOML OVERVIEW

AutoML is an automated machine learning system in Red Hat OpenShift AI that finds the best model for your prediction task. You provide training data in CSV format and select a task type. AutoML trains and evaluates multiple models, ranks them on a leaderboard, and produces notebooks that you can use to run predictions with the best-performing model.

1.1. AUTOML WORKFLOW

When you create an AutoML optimization run, AutoML loads your training data, samples it if needed, splits it into training and test sets, and trains candidate models by using different algorithms and configurations. AutoML evaluates each model against the held-out test set. The leaderboard ranks models by the optimized metric for your task type, and you can sort by other metrics to compare performance. You can register a model to a model registry or save a notebook for evaluation and exploration. After you register a model, you can deploy it for inference with a compatible serving runtime.

1.2. SUPPORTED TASK TYPES

Binary classification

Predict outcomes with two distinct categories, such as pass or fail, or approved or denied.

Multiclass classification

Predict outcomes with three or more distinct categories, such as product categories or support ticket priorities.

Regression

Predict continuous numerical values, such as price, temperature, or duration.

Time series forecasting

Predict future values over a specified date or time range. Your data must include a timestamp column, a numeric target column, and an ID column that identifies each time series.

1.3. TECHNOLOGY PREVIEW LIMITATIONS

The following limitations apply during Technology Preview:

- CSV format training data only
- Training data capped at 32 MiB when uploaded through the dashboard, or 100 MB when loaded from S3
- No custom algorithm selection or hyperparameter configuration
- Optimization runs cannot be edited after creation

1.4. VIEWING EXTERNALLY CREATED RUNS

If you import an AutoML pipeline to your pipeline server and create runs from it, the runs appear on the AutoML page in the dashboard. When you import the pipeline, set the pipeline name and version name according to the task type:

- For tabular tasks such as binary classification, multiclass classification, and regression, set the pipeline name to **autogluon-tabular-training-pipeline** and the pipeline version name to

autogluon-tabular-training-pipeline-`<version>`, such as **autogluon-tabular-training-pipeline-3.4.0**.

- For time series forecasting, set the pipeline name to **autogluon-timeseries-training-pipeline** and the pipeline version name to **autogluon-timeseries-training-pipeline-`<version>`**, such as **autogluon-timeseries-training-pipeline-3.4.0**.

For more information about importing pipelines, see [Importing a pipeline](#).

Additional resources

- [Create an AutoML optimization run](#)
- [Evaluate AutoML results](#)
- [Run predictions with an AutoML model](#)
- [Deploy an AutoML model for inference](#)
- [AutoML evaluation metrics](#)
- [AutoML configuration parameters](#)

To find the best model for your data, create an AutoML optimization run, evaluate the results, and register or test the best-performing model.

Before you begin, ensure that the following prerequisites are met:

- You have access to a data science project with an AI Pipelines server configured. For more information, see [Configuring a pipeline server](#).
- Your training data is available in an S3-compatible storage bucket in CSV format.

CHAPTER 2. CREATE AN AUTOML OPTIMIZATION RUN

Create an AutoML optimization run to train and evaluate machine learning models for your data set.

Prerequisites

- You have editor access to a project in OpenShift AI.
- You have access to a data science project with an AI Pipelines server configured.
- Your cluster has at least 4 CPUs and 16 GiB memory available for scheduling.
- Your training data is available in an S3-compatible storage bucket in CSV format.
- Your CSV file uses UTF-8 encoding and comma delimiters, and includes a header row. Maximum file size is 32 MiB when uploading through the dashboard.

Procedure

1. In the OpenShift AI dashboard, click **Develop and train > AutoML**.
2. Select a project and click **Create AutoML optimization run**.
3. Enter a name and optionally a description for the optimization run and click **Next**.
4. Configure the data source as follows:
 - a. From the **S3 connection** list, select an existing data connection or click **Add new connection**.
 - b. Click **Browse bucket** and select your CSV data file.
5. Select the prediction task type and configure the prediction settings:
 - **Binary classification:** Your target column has two distinct categories.
 - **Multiclass classification:** Your target column has three or more distinct categories.
 - **Regression:** Your target column has continuous numerical values.
 - **Time series forecasting:** Your data is sequential and you want to predict future values.
 - a. For binary classification, multiclass classification, or regression:
 - Select the **Label column** from your data set.
 - b. For time series forecasting:
 - Select the **Target column** with the numerical values to forecast.
 - Select the **Timestamp column** with the date or time values.
 - Select the **ID column** that identifies each time series.
 - Optional: Select **Known covariates** for values known in advance across the forecast horizon.

- Set the **Prediction length** for the number of time steps to forecast.
6. Optional: Adjust the **Top models to consider** value (default 3, range 1 to 10 for tabular, 1 to 7 for time series).

**NOTE**

Optimization runs cannot be edited after creation. To stop, archive, or delete the underlying pipeline run, see [Managing pipeline runs](#).

7. Click **Create run**.
AutoML begins training models. You can monitor the run status on the AutoML page.

Verification

- On the **AutoML** page, verify that the optimization run is displayed in the list with a status indicating that it is running or complete.

Additional resources

- [Evaluate AutoML results](#)
- [AutoML configuration parameters](#)

CHAPTER 3. EVALUATE AUTOML RESULTS

After an AutoML optimization run completes, review the leaderboard and model details to select the best model for your use case. You can register a model to a model registry or save a notebook for evaluation and exploration.

Prerequisites

- You have created an AutoML optimization run and the run status is **Completed**. For more information, see [Create an AutoML optimization run](#).

Procedure

1. In the OpenShift AI dashboard, click **Develop and train > AutoML**.
2. Click the name of your completed run to view the results.
3. Review the leaderboard, which ranks trained models by the optimized metric for your task type. The top-ranked model is highlighted. Click a column header to sort by a different metric.
4. To view detailed information about a model, click **View details** from the actions menu. The model details view includes evaluation metrics, feature importance scores, and a confusion matrix. Feature importance is not available for time series forecasting tasks. The confusion matrix is available for binary and multiclass classification tasks only.
5. Review the feature importance and the confusion matrix across models to verify that the best-scoring model produces predictions that you can explain and trust.
6. To register the selected model to a model registry, select **Register model** from the actions menu and select a model registry.
7. To save a notebook, select **Save notebook** from the actions menu.

Verification

- If you registered a model, navigate to the model registry and verify that the model is displayed.
- If you saved a notebook, verify that the notebook file downloaded to your local system.

Additional resources

- [Run predictions with an AutoML model](#)
- [Deploy an AutoML model for inference](#)
- [AutoML evaluation metrics](#)
- [Working with model registries](#)

CHAPTER 4. RUN PREDICTIONS WITH AN AUTOML MODEL

After you save a notebook from the AutoML leaderboard in OpenShift AI, run it in a workbench to load the trained model and run predictions on sample data.

Prerequisites

- You have created an AutoML optimization run and saved a notebook from the leaderboard. For more information, see [Evaluate AutoML results](#).
- You have a running workbench in your OpenShift AI project.
- You have the connection details for the S3-compatible object storage bucket configured for the AI Pipelines server in your project.

Procedure

1. In the OpenShift AI dashboard, open your workbench.
2. Add the S3-compatible object storage connection configured for the AI Pipelines server to the workbench:
 - a. On the workbench edit page, scroll to the **Connections** section.
 - b. Click **Attach existing connections**.
 - c. Select the connection and click **Attach**.
 - d. Click **Update workbench**.
3. Upload the saved notebook to the workbench.
4. Open the notebook and run all cells. The notebook loads the trained model from S3 and runs predictions on sample data.
The final cells in the notebook display sample predictions from test inputs.

Verification

- Verify that all cells in the notebook completed without errors.
- Verify that the model returns predictions for test inputs.

Additional resources

- [Evaluate AutoML results](#)

CHAPTER 5. DEPLOY AN AUTOML MODEL FOR INFERENCE

After you register a model from the AutoML leaderboard, you can deploy it for inference with the AutoGluon serving runtime. Deploying the model creates a REST API endpoint that you can use to send prediction requests.

Prerequisites

- You have registered a model from the AutoML leaderboard to a model registry. For more information, see [Evaluate AutoML results](#).
- An administrator has added the AutoGluon serving runtime to your cluster by following the steps in [Adding a custom model-serving runtime](#) with the following **ServingRuntime** definition:

Example ServingRuntime with AutoGluon

```
apiVersion: serving.kserve.io/v1alpha1
kind: ServingRuntime
metadata:
  name: kserve-autogluonserver
  annotations:
    openshift.io/display-name: "AutoGluon ServingRuntime for KServe"
spec:
  annotations:
    prometheus.kserve.io/port: "8080"
    prometheus.kserve.io/path: "/metrics"
  supportedModelFormats:
    - name: autogluon
      version: "1"
  protocolVersions:
    - v1
    - v2
  containers:
    - name: kserve-container
      image: registry.redhat.io/rhoai/odh-kserve-autogluon-server-
        rhel9@sha256:6d55374ef2e2ac09d772701a72d2ea370c57c34eb89c7b7a71ee204559c699a3

    args:
      - --model_name={{.Name}}
      - --model_dir=/mnt/models
      - --http_port=8080
  securityContext:
    allowPrivilegeEscalation: false
    privileged: false
    runAsNonRoot: true
  capabilities:
    drop:
      - ALL
  resources:
    requests:
      cpu: "1"
      memory: 2Gi
    limits:
      cpu: "1"
      memory: 2Gi
```

Procedure

1. Deploy the registered model version from the model registry by following the steps in [Deploying a model version from a model registry](#).
Configure the deployment with the following values:

- For **Model framework**, select **autogluon - 1**.
- For **Serving runtime**, select **AutoGluon ServingRuntime for KServe**.



IMPORTANT

If you are deploying a time series forecasting model and your training data used column names other than **item_id** and **timestamp**, you must add custom runtime environment variables before deploying. See the following table.

Table 5.1. Custom runtime environment variables for time series models with non-default column names

Environment variable	Description
AUTOGLUON_TS_ID_COLUMN	The name of the column that identifies each time series. Set this variable if the column is not named item_id .
AUTOGLUON_TS_TIMESTAMP_COLUMN	The name of the column that has the date or time values. Set this variable if the column is not named timestamp .

Add these variables under **Advanced settings** with the same column names that you specified when you created the optimization run.

Verification

- On the **Deployments** page, verify that the model deployment shows a status of **Ready**.

Additional resources

- [Evaluate AutoML results](#)
- [Adding a custom model-serving runtime](#)
- [Deploying a model version from a model registry](#)

CHAPTER 6. AUTOML EVALUATION METRICS

AutoML evaluates each trained model with metrics appropriate for the prediction task type. The leaderboard ranks models by the optimized metric. Additional metrics might appear on the leaderboard depending on the task type.

6.1. OPTIMIZED METRICS BY TASK TYPE

AutoML selects the optimization metric based on your task type.

Table 6.1. Optimized metrics

Task type	Optimized metric
Binary classification	Accuracy
Multiclass classification	Accuracy
Regression	R^2
Time series forecasting	MASE (Mean Absolute Scaled Error)

Accuracy

The proportion of predictions that are correct. A high accuracy score means the model correctly classifies most inputs.

R^2 (R-squared)

The proportion of variance in the target variable that the model explains. A score of 1.0 means the model perfectly predicts the target. A score of 0.0 means the model performs no better than predicting the mean.

MASE (Mean Absolute Scaled Error)

A scale-independent measure of forecast accuracy. AutoML negates MASE values so that higher values indicate better models, consistent with all other metrics on the leaderboard. A raw MASE below 1.0 means the model outperforms a naive baseline forecast.

6.2. MODEL DETAIL VIEWS

Feature importance

Shows which input features had the most influence on the model's predictions. Use feature importance to verify that the model relies on meaningful features rather than noise or data artifacts. Feature importance is available for binary classification, multiclass classification, and regression tasks. It is not available for time series forecasting tasks.

Confusion matrix

Shows predicted compared to actual class distributions as a grid. Use the confusion matrix to identify which classes the model confuses most often. The confusion matrix is available for binary and multiclass classification tasks only.

Additional resources

- [Evaluate AutoML results](#)

- [AutoML configuration parameters](#)

CHAPTER 7. AUTOML CONFIGURATION PARAMETERS

You can configure several parameters when you create an AutoML optimization run in the OpenShift AI dashboard.

7.1. COMMON PARAMETERS

You set the following parameters for all task types.

Table 7.1. Common parameters

Parameter	Description	Values
S3 connection	The data connection for your S3-compatible storage bucket.	Existing data connections in your project, or create a new connection.
Training data file	The CSV file containing your training data. Must use UTF-8 encoding and comma delimiters with a header row.	CSV format. Maximum 32 MiB when uploading through the dashboard, or 100 MB when loaded from S3.
Prediction task type	The type of prediction problem.	Binary classification, Multiclass classification, Regression, Time series forecasting
Top models to consider	The number of top-performing models to refit on the full training set.	1-10 (tabular) or 1-7 (time series), default 3

7.2. TABULAR TASK PARAMETERS

You set the following parameter for binary classification, multiclass classification, and regression tasks.

Table 7.2. Tabular task parameters

Parameter	Description	Values
Label column	The column in your data set that contains the values to predict.	Populated from the CSV file schema.

7.3. TIME SERIES TASK PARAMETERS

You set the following parameters for time series forecasting tasks.

Table 7.3. Time series task parameters

Parameter	Description	Values
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Parameter	Description	Values
Target column	The column with the numerical values to forecast.	Populated from the CSV file schema.
Timestamp column	The column with the date or time values.	Populated from the CSV file schema.
ID column	The column that identifies each time series.	Required. Populated from the CSV file schema.
Known covariates	Columns with values that are known in advance across the forecast horizon.	Optional. Populated from the CSV file schema.
Prediction length	The number of time steps to forecast.	Positive integer.

7.4. AUTO-SELECTED PARAMETERS

AutoML automatically selects the following during training. You cannot configure these values.

- Algorithms and hyperparameters: AutoML uses AutoGluon to select and tune candidate models.
- Train/test split: AutoML splits your data into training and test sets for evaluation.
- Evaluation metric: AutoML selects the optimization metric based on your task type. For more information, see [AutoML evaluation metrics](#).

Additional resources

- [Create an AutoML optimization run](#)
- [AutoML evaluation metrics](#)